Kushal Mamillapalli

km6238@nyu.edu | linkedin.com/in/kushal-mamillapalli | github.com/Techdude01 | Secaucus, NJ

Education

New York University — Brooklyn, NY
B.S. in Computer Science, Minor in Mathematics | GPA: 3.9/4.0, Dean's List — *Sep. 2023 - May 2026*

- Coursework: Data Structures, Algorithms, Operating Systems, Databases, Machine Learning, Data Science.

Technical Skills

Languages: Scala, Java, Python, C++, Bash, sbt, Maven
Data & Streaming: Apache Beam/Scio, Google Cloud Dataflow, BigQuery (SQL), Bigtable, Pub/Sub, Protobuf, Avro
ML & Vision: ROS 2, TensorRT, DeepStream, CUDA/NVMM, PyTorch, YOLOv8, scikit-learn, Prophet, Optuna
Cloud Infrastructure: Google Cloud IAM, Kubernetes, GitHub Actions/CI
Certifications: AWS Certified Machine Learning Engineer - Associate (May 2026)

Experience

Spotify — Jun. 2026 – Aug. 2026
Data Engineering Intern — *New York City, NY*

- For Spotify's batch/streaming events supporting ML and user data, built a Claude skill using 5 MCPs to streamline event-deprecation and data-parity checks, cutting migration setup time an estimated 35%.
- Root-caused and fixed 4 shared Python CLI defects (serialization, API payloads, timeouts, response handling) and restored data validation, unblocking all engineers.
- Consolidated 3 legacy events into a canonical Protobuf event schema across 5 batch/streaming repositories, migrated consumers, and validated 4,560/4,560 production records in BigQuery.
- Designed an experiment proving Protobuf renames with unchanged event IDs and field numbers preserve outputs across four batch, streaming, and SQL read paths, de-risking ∼16 planned renames.

NYU Enterprise Data Management — Apr. 2025 – Present
Technical Systems Engineering Intern (Part-Time) — *New York City, NY*

- Reduced runtime 40% for NYU Financial Aid's ETL pipeline by optimizing SQL/Pandas and migrating 100GB+ inputs to columnar Parquet, reducing storage and improving read times 54%.
- Developed a 600K-row SARIMA/Prophet forecasting engine with 4% MAPE, visualizing departmental trends on an Oracle Analytics Cloud dashboard.
- Established the team's first GitHub version-control workflow and removed 100+ legacy files and secrets from the server, improving security and cutting debugging time 75%.

NYU ARC Robotics — Jan. 2024 – Jun. 2026
Computer Vision Lead — *New York City, NY*

- Engineered the NYU ARC Robotics League robot's C++ ROS 2 auto-aim system with Basler/RealSense, YOLO, and TensorRT/DeepStream.
- Reduced latency 70% (40ms to 12ms) and raised throughput 2.3x (∼67 to 156 FPS) with CUDA/NVMM.
- Boosted armor-targeting F1 15% with tracking, 3D PnP/ballistics, and STM32 UART commands.
- Containerized Jetson vision stack with Docker/systemd, health checks, model selection, and tooling.

Projects

NBAnomaly | GitHub — Dec. 2025 – Present

- Developed a general-use NBA platform to surface player under- and overperformance across 10+ seasons with FastAPI on AWS EC2 and resilient ingestion.
- Validated 78% backtest precision with an Isolation Forest over 15+ engineered features.
- Lowered LLM report latency from ∼4.5s to 2.5ms with a cache-aside PostgreSQL layer.

AutoCPT (HackNYU 2025 Winner) | GitHub — Feb. 2025

- Created a live-visit assistant that turns clinician-patient voice conversations into suggested CPT codes, saving clinicians 16+ hours/week with Whisper, LLaMA-3, and Groq.
- Enabled <500ms live responses with async Flask, retry/audit logic, and React YOLOv8 overlays.